

Lecture-21
Mathematics 1 (15B11MA111)

CO [C105.4]

Module: Vector Calculus

Topics to be covered

➤ Green's Theorem

- ❖ **Proof**

- ❖ **Verification**

- ❖ **Application**

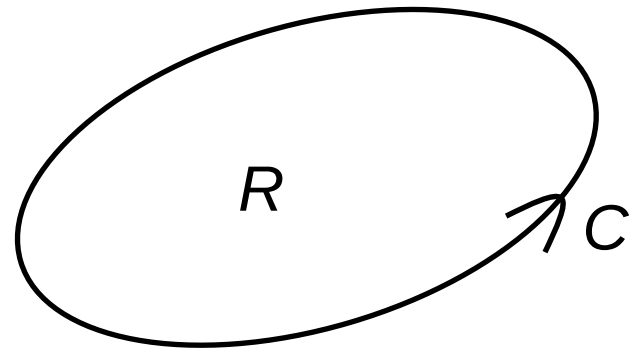
➤ Assignments

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Green's Theorem:

It provides a relationship between a double integral over a region R and the line integral over the closed curve C bounding R . Green's theorem is also called the first fundamental theorem of integral vector calculus.



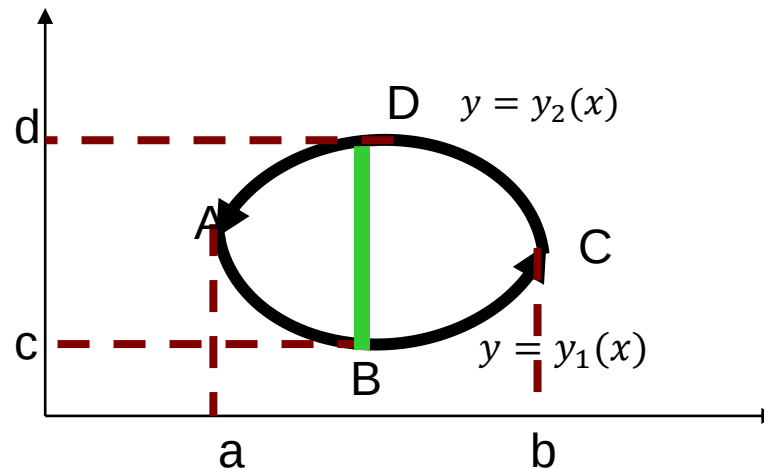
Let C be a closed curve bounding the region R . If $f, g, \frac{\partial f}{\partial y}$ and $\frac{\partial g}{\partial x}$ are continuous on R , then

$$\oint_C (f(x, y)dx + g(x, y)dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dxdy$$

the integration being carried in the positive direction (*counter clockwise direction*) of C .

Proof:

Let the curve C be divided into two curves $C_1(ABC)$ & $C_2(CDA)$.



$$\iint_R \frac{\partial f}{\partial y} dx dy = \int_{x=a}^{x=b} \left[\int_{y=y_1(x)}^{y=y_2(x)} \frac{\partial f}{\partial y} dy \right] dx$$

$$\begin{aligned}
&= \int_a^b [f(x, y_2) - f(x, y_1)] dx \\
&= \int_a^b [f(x, y_2) - f(x, y_1)] dx \\
&= - \int_b^a f(x, y_2) dx - \int_a^b f(x, y_1) dx \\
&= - \left[\int_b^a f(x, y_2) dx + \int_a^b f(x, y_1) dx \right]
\end{aligned}$$

$$\begin{aligned}
&= - \left[\int_{C_2} f(x, y) dx + \int_{C_1} f(x, y) dx \right] \\
&= - \oint_C f(x, y) dx
\end{aligned}$$

Thus

$$\oint_C f(x, y) dx = - \iint_R \frac{\partial f}{\partial y} dx dy$$

Similarly , we have

$$\oint_C g dy = \iint_R \frac{\partial g}{\partial x} dx dy$$

Combining both

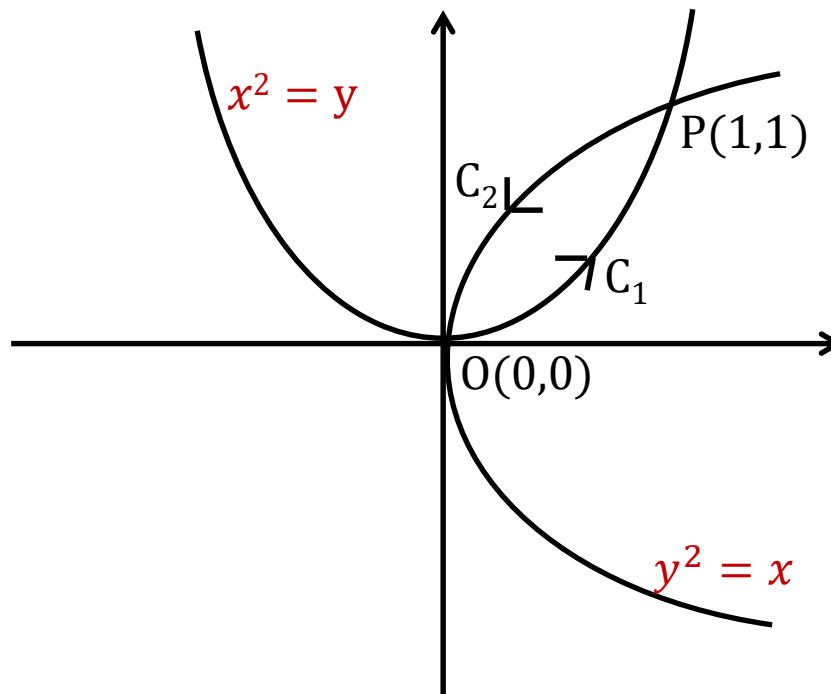
$$\oint_C (f dx + g dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$

$$\oint_C (f(x, y) dx + g(x, y) dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$

Example: Using Green's theorem evaluate $\oint_C ((x^2 + y^2)dx + (y + 2x)dy)$, where C is the boundary of the region in the first quadrant that is bounded by the curves $y^2 = x$ and $x^2 = y$.

Solution: Green's theorem

$$\oint_C (f(x, y)dx + g(x, y)dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$



Solution cont....: Green's theorem

$$\oint_C (f(x,y)dx + g(x,y)dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dxdy$$

$$= \iint_R (2 - 2y) dxdy$$

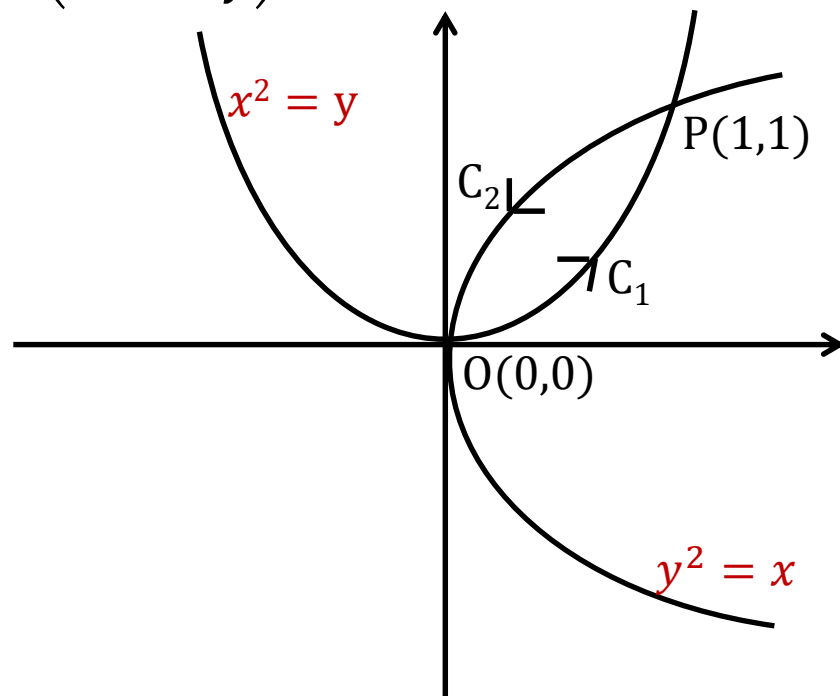
$$= \int_0^1 \left[\int_{x^2}^{\sqrt{x}} (2 - 2y) dy \right] dx$$

$$= \int_0^1 [2y - y^2]_{x^2}^{\sqrt{x}} dx = \dots$$

$$= \int_0^1 (2\sqrt{x} - x - 2x^2 + x^4) dx$$

$$= \dots$$

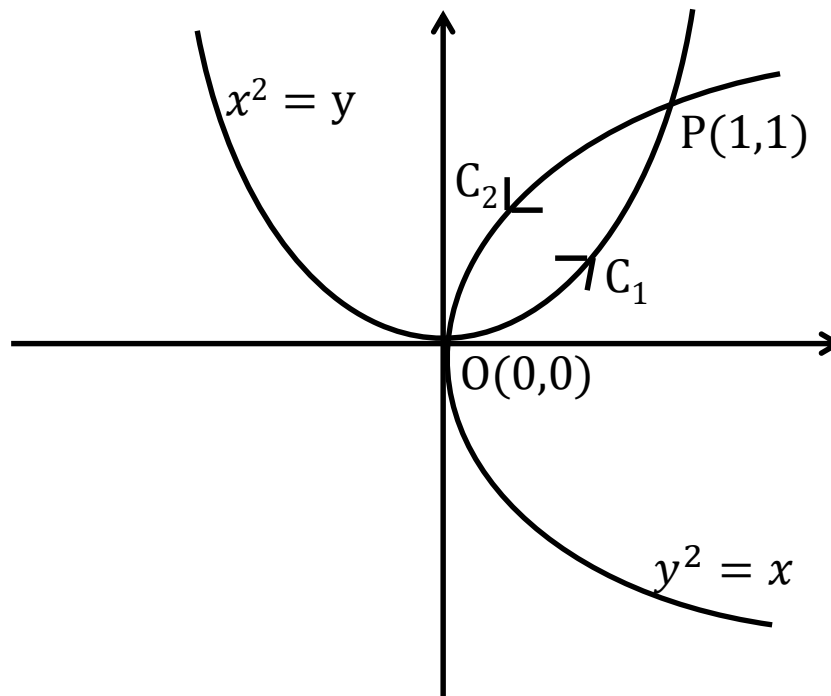
$$= \dots = \frac{11}{30}$$



Example: Verify the Green's theorem for $\oint_C ((x^2 + y^2)dx + (y + 2x)dy)$, where C is the boundary of the region in the first quadrant that is bounded by the curves $y^2 = x$ and $x^2 = y$.

Solution: Green's theorem

$$\oint_C (f(x, y)dx + g(x, y)dy) = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dxdy$$



$$\mathbf{RHS} = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$

Same as previous example... $=\frac{11}{30}$

$$\begin{aligned} \mathbf{LHS} &= \int_C (f(x, y)dx + g(x, y)dy) \\ &= \left[\int_{C_1} + \int_{C_2} \right] (f dx + g dy) \end{aligned}$$

$$= I_1 + I_2$$

We have along curve C_1 $x^2 = y$

Let $x = t$ so that $y = t^2$

hence $dx = dt$ and $dy = 2t dt$

$$I_1 = \int_{C_1} ((x^2 + y^2)dx + (y + 2x)dy)$$

$$= \int_0^1 ((t^2 + t^4)dt + (t^2 + 2t)2t dt)$$

$$= \int_0^1 (t^2 + t^4 + 2t^3 + 4t^2)dt$$

$$= \int_0^1 (5t^2 + 2t^3 + t^4)dt$$

$$= \left[5 \frac{t^3}{3} + \frac{t^4}{2} + \frac{t^5}{5} \right]_0^1$$

$$= \frac{71}{30}$$

We have along curve C_2 $x = y^2$

Let $y = t$ so that $x = t^2$

hence $dy = dt$ and $dx = 2t dt$

$$\begin{aligned} I_2 &= \int_{C_2} ((x^2 + y^2)dx + (y + 2x)dy) \\ &= \int_1^0 ((t^4 + t^2)2t dt + (t + 2t^2)dt) \\ &= \int_1^0 (2t^5 + 2t^3 + t + 2t^2)dt = \dots = -2 \end{aligned}$$

$$\text{Hence LHS} = \frac{71}{30} - 2 = \frac{11}{30} = \text{RHS}$$

Example: Using Green's theorem evaluate $\oint_C (x^2 y dx + x^2 dy)$ where C is the boundary described counter clockwise of the triangle with vertices (0,0), (1,0), (1,1).

Solution:

$$\begin{aligned}\oint_C (f dx + g dy) &= \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy \\ &= \oint_C (x^2 y dx + x^2 dy) \\ &= \iint_R (2x - x^2) dx dy \\ &= \int_0^1 (2x - x^2) dx \int_0^x dy \\ &= \int_0^1 (2x^2 - x^3) dx \\ &= \frac{5}{12}\end{aligned}$$

Assignment

1. Verify the Green's Theorem for $\oint_C (xy^2 + 2xy)dx + 5x^3dy$, where C is the boundary of the region enclosing $y^2 = 4x, x = 3$.
2. Using the Green's Theorem evaluate $\oint_C (xy^2dx + x^2ydy)$, where C is the rectangle with vertices at $(-1, 0), (2, 0), (2, 2)$ and $(-1, 2)$. **Ans 0.**

Thank you